Control: Quadrotor Control Intro & Modern Control Design

Course notes – Autumn 2022/23

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Course: UAVs, MEAer (Técnico Lisboa), Autumn Semester 2022/2023. Instructors: Rita Cunha / J.R. Azinheira. Ch4's first part follows "Modern Control Design" lecture notes by Pedro Batista (Feb 2018); the LQR/pole-placement worked example is adapted from J. Hespanha, *Linear Systems Theory*, Princeton Univ. Press, 2009 (aircraft roll dynamics example).

1 Chapter 3 - Quadrotor Control Introduction

1.1 Starting point: the quadrotor dynamic model

From Ch1/Ch2 (rigid body + quadrotor modeling), the equations of motion used throughout are:

- Translational dynamics: $m\ddot{p} = -Tr3 + mge3$, with $r3 = Re3$ (third column of the rotation matrix, i.e. the body z-axis expressed in the inertial frame).
- Attitude kinematics: $\dot{R} = RS(\omega)$ (S(.) = skew-symmetric operator, so $S(\omega)v = \omega x v$).
- Rotational dynamics: $J\dot{\omega} = -S(\omega)J\omega + n$.

Inputs to the system are total thrust `T` (scalar) and torque `n` (3-vector). Outputs of interest are position/velocity `(p, p_dot)` and attitude/angular velocity `(R, omega)`.

Small-angle/near-hover exercise: linearizing `p_ddot` near hover with roll `phi` and pitch `theta` gives $\ddot{p} \approx [-g\theta, g\phi, 0]^T$ — this is the classical result that pitch controls x-acceleration and roll controls y-acceleration (sign convention as defined by the course's rotation order).

1.2 Control objective and hierarchical (cascade) structure

Objective: **trajectory tracking**. Given a desired flat-output trajectory $(p(t), \psi(t))$ (position + yaw), find `(T, n)` such that $(p(t), \psi(t)) \to (p(t), \psi(t))$ as $t \to \infty$.

Because of the block-diagram dependency structure (rotational dynamics feed into translational dynamics via `r3`), the natural design is **hierarchical/cascade control**:

- **Outer loop (position controller)**: takes $(p, \dot{p}, \ddot{p})$ and `(p, p_dot)`, produces desired thrust `T` and a desired direction `r3d` for the body z-axis. Goal: track position.
- **Inner loop (attitude controller)**: takes the desired attitude (derived from `r3d` and desired yaw ψ) and current `(R, omega)`, produces torque `n`. Goal: track attitude. Runs faster (5-10x bandwidth) than the outer loop, which is the standard justification for treating the loops as (approximately) independent SISO problems.

Design recipe (4 steps):

1. **Neglect rotational dynamics**: replace `r3` by a desired `r3d`, define virtual input $u_T = -(T/m)r3d$. Then $T = m\|u_T\|$ and $r3d = -u_T/\|u_T\|$ (T is magnitude, r3d is direction).
2. **Design a control law for `u_T`** assuming $\ddot{p} = u_T + ge3$ (or $= u_T + g$ in the slide's shorthand).
3. **Combine `r3d` with** ψ to get a full desired attitude `R_d` (or Euler-angle equivalent $lambda_d = (phi_d, theta_d, \psi)$). One explicit formula given: $r3d = Rz(\psi)Ry(theta_d)Rx(phi_d)e3$, leading to $[\cos(phi_d)\sin(theta_d), -\sin(phi_d), \cos(phi_d)\cos(theta_d)]^T = Rz(-\psi)r3d$ — i.e. invert this to solve for `(phi_d, theta_d)` given `r3d` and ψ.

4. **Design a control law for `n`** to track `R_d/lambda_d`, using either the rotation-matrix kinematics $\dot{R} = RS(\omega)$ or the Euler-angle form $\dot{\lambda} = Q(\phi, \theta)\omega$ together with $J\dot{\omega} = -S(\omega)J\omega + n$.

1.3 Position (translational) controller design

Model: $\ddot{p} = u_T + g$ (per-axis double integrator with gravity offset), reference $(p(t), \dot{p}(t), \ddot{p}(t))$.

Error variables: $e = p - p$, $\dot{e} = \dot{p} - \dot{p}$, giving $\ddot{e} = u_T + g - \ddot{p}$.

PD control law: $u_T(t) = -kPe(t) - kD\dot{e}(t) - g + \ddot{p}(t)$, which yields the clean closed-loop error dynamics $\ddot{e}(t) = -kPe(t) - kD\dot{e}(t)$ (feedback term) $+\ddot{p}$ (feedforward term, cancels out in the error equation). This is a linear 2nd-order system:

- Stable (asymptotically) iff $kP > 0$ and $kD > 0$ — `kP` acts like a spring, `kD` like a damper.
- Standard 2nd order form comparison: closed-loop transfer function $P(s)/P(s) = (kP + kDs)/(s^2 + kDs + kP)$, compared against $G(s) = wn^2/(s^2 + 2\xi wns + wn^2)$.
- Damping ratio: $\xi = kD/(2\sqrt{kP})$. Increasing `kD` increases `xi` (less overshoot); increasing `kP` decreases `xi`. Settling time improves as `kD` increases (ξwn increases).

1.4 Classical SISO analysis tools used

- **Root locus**: rewrite $C(s) = kP + kDs = k(s + z)/z$ with $z = kP/kD$, $k = kP$, open-loop $G(s) = k(s + z)/(zs^2)$. Increasing gain `k` (with fixed zero `z`) pulls the poles further into the left half-plane -> faster, less oscillatory response (compare $k = 1, z = 1$ vs $k = 24, z = 2$ step responses).
- **Bode / loop-shaping**: with the open-loop $L(s) = k(s + z)/(zs^2)$, read off:
  - Gain margin `GM` and phase margin `PM` from the Bode plot (example: $k = 24, z = 2$ gives $GM = +inf$, $PM = 80.7°$ at crossover $wc = 12.2 rad/s$).
  - Maximum tolerable pure time delay from phase margin: $\tau < tau_c = PM/wc$ (numerically $0.116s$ in the example).
  - **Noise attenuation** spec: require $|Y(jwn)/N(jwn)|_dB \leq -10dB$ for `wn` in a high-frequency band (e.g. $[100, 1000]rad/s$), i.e. the loop gain must roll off enough at high frequency.
  - **Disturbance attenuation** spec: require $|Y(jwd)/D(jwd)|_dB \leq -80dB$ for low-frequency disturbances `wd` in e.g. $[0.01, 0.1]rad/s$, i.e. $|G(jwd)|_dB \geq 80dB$ (high loop gain at low frequency).
  - **Reference-following** spec similarly requires $|G(jwr)|_dB \geq 100dB$ for the reference-frequency band.

1.5 Disturbance rejection and integral action

Question posed: does the PD-controlled position loop reject a **constant wind disturbance** modeled as an acceleration input `w`? Using the final value theorem on $Y(s)/W(s) = 1/(s^2 + C(s))$ with $W(s) = w0/s$: **No** — steady-state error

$y(t) \to r(t) + w0/k$ (a constant offset proportional to disturbance magnitude and inversely proportional to gain $k = kP$).

Fix: add integral action -> PID controller. Strategy shown as adding a zero and an integrator to go from PD to PID: $C_PD(s) = k(s+z)/z$ -> $C_PID(s) = k(s+z)(s+z1)/(zz1s)$, equivalently $C(s) = kP + kI/s + kDs = (kDs^2 + kPs + kI)/s$. With integral action, $Y(s)/W(s) = s/(s^2 + kDs^2 + kPs + kI)$ (note the extra s in the numerator from the integrator), so $limy(t) = limsY(s) = 0$ for a step disturbance — the constant wind offset is rejected. Design guidance: keep the existing zero at -z, add the new integrator/zero pair at -z1 (example: $k = 16, z = 2, z1 = 0.5$).

Exercise left in the slides (unsolved): repeat the wind-rejection analysis for a **velocity** disturbance input instead of acceleration.

An experimental result slide shows a real quadrotor holding position in front of a desk fan (physical demonstration of disturbance rejection).

1.6 Extending hierarchical control to full state feedback (bridge to Ch4)

- Adding integral action to the position loop is generalized as **state feedback with an augmented integrator state**: $u_T(t) = -kP(p-p) - kD(\dot{p}-\dot{p}) - kIIntegral(p-p) - ge3 + \ddot{p}(t)$.
- The **attitude controller** is derived by linearizing about hover: equilibrium $lambda_0 = [0, 0, \psi]^T$, $omega_0 = 0$, $n_0 = 0$. Standard linearization $delta_xdot = (df/dx)|_eqdelta_x + (df/du)|_eqdelta_u$ applied to $\dot{\lambda} = Q(\phi, \theta)\omega$, $J\dot{\omega} = -S(\omega)J\omega + n$ gives, when J is diagonal, a decoupled double-integrator per axis: $delta_l\dot{a}mbda = delta_omega$, $Jdelta_\dot{o}mega = delta_n$. A **PD controller** is then used: $n = delta_n = -k1(\lambda - lambda_d) - k2\omega$.
- **Full hierarchical block diagram** (position PID -> attitude PD -> rotational dynamics -> translational dynamics) is reduced, for design purposes, to a simplified nested SISO model: outer loop $kp + kvs$ around an inner loop ($k_lambda + k_omegas$) feeding two integrators $1/s1/s$ (attitude+angular rate) into the position double integrator $1/s^2$. Design rule of thumb: make the inner loop 5-10x faster (higher bandwidth) than the outer loop so they can be tuned quasi-independently. Root-locus examples compare $C1(s) = 5(s+1)$ inner vs $C2(s) = 10(s+2.5)$ outer (well-separated, clean step response) against $C2(s) = 3(s+2.5)$ (insufficient separation -> oscillatory response).
- **Full pole-placement version (last slides, bridging into Ch4's state-space viewpoint)**: model the whole cascade (position PID error states + inner-loop dynamics) as one augmented state-space system and place poles directly with a single gain vector K, e.g. $K = place(A2, B2, [-0.5+0.2i, -0.5-0.2i, -0.1, roots([1, k_omega, k_lambda])])$, giving a 5-element state feedback gain $u = theta_r = [0.029, 0.419, 1.4973, 1.1975, 0.275][x_I; y-r; ydot-rdot; \theta; thetadot]$. Explicitly noted: **this is no longer hierarchical** — it's a single full-state feedback law over the combined state, which is exactly the "modern control design" (state-space) approach developed in Ch4.

2 Chapter 4 - Modern Control Design

Two parts: (1) general linear state-space control/estimation theory (controllability, observability, pole placement, observers, LQR), following P. Batista's notes; (2) a worked LQR design example on aircraft roll dynamics (from Hespanha's textbook), connecting LQR back to loop-shaping/Bode ideas used in Ch3.

2.1 Linear state-space models

General (possibly time-varying) form: $\dot{x}(t) = A(t)x(t) + B(t)u(t)$, $y(t) = C(t)x(t) + D(t)u(t)$. The course always sets $D(t) = 0$ (no direct feedthrough). When `A, B, C` are constant -> **LTI system**: $\dot{x} = Ax + Bu$, $y = Cx$.

- **Transfer function matrix** (LTI only): Laplace transform with zero initial conditions gives $Y(s) = C(sI - A)^{-}1BU(s)$, i.e. $G(s) = C(sI - A)^{-}1B$.
- **Time-domain solution** (LTI, initial state `x0` at `t0`): $x(t) = e^{A(t-t0)}x0 + Integral_{t0}^{t} e^{A(t-\sigma)}Bu(\sigma)dsigma$, and correspondingly for `y(t)`.

2.2 Controllability

Definition: the system is controllable on `[t0, tf]` if for any `x0`, there exists a continuous input `u(t)` driving $x(tf) = 0$ (note the slide's convention: driving state to **zero**, not to an arbitrary target — equivalent statement for LTI systems since the reachable set is a subspace).

Theorem (LTI): the system is controllable iff the controllability matrix $C := [B, AB, A^2B, ..., A^{n-1}B]$ has full rank `n` (n = number of states). Worked 2-state examples: $A = [[0, 1], [0, 0]]$, $B = [0, 1]$ (i.e. $xdot1 = x2, xdot2 = u$) is controllable (`C` full rank); $A = [[0, 1], [0, 0]]$, $B = [1, 0]$ (i.e. $xdot1 = x2 + u, xdot2 = 0$) is **not** controllable (rank-deficient `C`) — intuition: `x2` is never affected by `u`, so it cannot be driven.

2.3 Observability

Definition: observable on `[t0,tf]` if the initial state `x0` is uniquely determined from `y(t)` for `t` in `[t0,tf]`. Explicitly noted: for **known-input** LTV systems, observability does not depend on whether `u` is zero or not (subtract the known-input contribution from the output first) — for nonlinear systems this does NOT hold in general (much harder analysis).

Theorem (LTI): observable iff the observability matrix $O := [C; CA; ...; CA^{n-1}]$ has full rank `n`. Worked examples mirror controllability: $C = [1, 0]$ on the same `A` is observable; $C = [0, 1]$ is not (can't tell `x1` apart from the output).

2.4 State feedback and pole placement

Linear state feedback: $u(t) = r(t) - Kx(t)$ (block diagram: reference `r`, minus `Kx` feedback, into $xdot = Ax + Bu$).

Pole placement theorem: if `(A,B)` is controllable, then for **any** desired real monic degree-n polynomial `p(lambda)`, there exists a constant gain `K` such that $\det(\lambda I - A + BK) = p(\lambda)$ — i.e. the closed-loop eigenvalues can be placed arbitrarily.

2.5 State observers (Luenberger observer)

Motivation: state feedback needs the full state, which usually isn't measured directly -> build a state estimate.

Luenberger observer: $x\dot{hat}(t) = A(t)xhat(t)+B(t)u(t)+L(t)[y(t)-C(t)xhat(t)]$ — a copy of the plant driven by the actual input, corrected by the output estimation error weighted by gain `L`.

Estimation error $xtilde := x - xhat$ obeys $xti\dot{l}de(t) = [A(t) - L(t)C(t)]xtilde(t)$ — autonomous linear dynamics independent of `u` and `y`; convergence requires `A-LC` to be stable (eigenvalues negative real part / Hurwitz).

Duality: eigenvalues of `A - LC` equal eigenvalues of $(A - LC)^T = A^T - C^T L^T$. This maps the observer design problem onto the pole-placement problem under the substitution $A <\rightarrow A^T$, $B <\rightarrow C^T$, $K <\rightarrow L^T$ — i.e. designing `L` for observability is mathematically identical to designing `K` for controllability, just transposed. Observer poles chosen via $L = place(A', C', pCL)'$ in the worked example (reusing the same closed-loop pole set as the LQR design, $pCL = eig(A - BK)$).

2.6 Combining state feedback + observer: separation principle

Using estimated state in the feedback law: $u(t) = r(t) - Kxhat(t)$. Substituting gives the augmented `[x; xtilde]` system: $[xdot; xtildedot] = [[A - BK, BK], [0, A - LC]][x; xtilde] + [B; 0]r(t)$, $y = [C, 0][x; xtilde]$.

Theorem/Separation principle: because the augmented system matrix is block upper-triangular, its eigenvalues are exactly the union of the eigenvalues of `A-BK` and `A-LC`. **Conclusion: controller gain `K` and observer gain `L` can be designed completely independently** by pole placement (or, more strongly, independently as *optimal* LQR/estimator problems — "more on this later" per the slides). This does **not** generalize to nonlinear systems (no general separation result).

2.7 Optimal control: Linear Quadratic Regulator (LQR)

Motivation: pole placement lets you "shape" closed-loop dynamics but doesn't optimize anything; LQR minimizes a cost.

LQR problem: for $xdot = Ax + Bu$, minimize $J = Integral_0^i nf[x^T(t)Qx(t) + u^T(t)Ru(t)]dt$, with `Q` positive semi-definite and `R` positive definite.

Theorem: if `(A,B)` is *stabilizable* and `(A,G)` is *detectable* (with $Q = G^T G$), then:

- There is a unique positive-definite solution `P` to the **algebraic Riccati equation (ARE)**: $A^T P + PA + Q - PBR^-1B^T P = 0$.
- The optimal feedback is $u(t) = -Kx(t)$ with $K := R^-1B^T P$, achieving minimum cost $J = x^T(0)Px(0)$.
- All closed-loop eigenvalues of `A - BK` have negative real part (guaranteed stability, not just optimality).

Stabilizability/detectability are **relaxations** of full controllability/observability: `(A,B)` stabilizable means only the *uncontrollable modes* need to be stable (don't need to control already-stable modes); `(A,G)` detectable means only the *unobservable modes* need to be stable.

Tuning `Q`/`R` — Bryson's rule (rule of thumb, "trial and error" per the slides): $Q_ii = 1/(max acceptable value of x_i^2)$, $R_ii = 1/(max acceptable value of u_i^2)$. Larger `Q` relative to `R` -> more control effort spent to keep the state small; larger `R` relative to `Q` -> conserve control effort at the cost of larger state excursions.

2.8 LQR worked example: aircraft roll dynamics (bridges LQR back to loop-shaping)

Linearized model (from Hespanha's book): $\dot{\phi} = \omega$, $\dot{\omega} = 0.8\omega - 20\tau$, $\dot{\tau} = -50\tau + 50u$; state $x = [\phi, \omega, \tau]$, output $y = \phi = Cx$ with $C = [1, 0, 0]$, $A = [[0, 1, 0], [0, -0.8, -20], [0, 0, -50]]$, $B = [0, 0, 50]^T$.

Cost: $J = Integral(x^T Qx + \rho u^2)dt$, with $Q = G^T G$, $G = [[1, 0, 0], [0, \gamma, 0]]$ (design weights $\rho > 0$, $\gamma > 0$), so the "regulated output" is $z(t) = Gx(t) = [\phi; \gamma\omega]$ — i.e. penalize roll angle and (scaled) roll rate, plus control effort weighted by `rho`.

Loop-shaping connection: define $P(s) = (sI - A)^-1B$ (plant, u to x), $L(s) = KP(s) = (sI - A)^-1BK$ (open-loop, scalar since SISO from `-u_bar` to `x` through `K`). Standard `GM`, `PM`, noise/disturbance/reference-tracking specs from Ch3 all carry over directly to this `L(s)`.

Kalman's equality (derivable from the ARE): $|1 + L(jw)|^2 = |1 + KP(jw)|^2 = 1 + \|GP(jw)\|^2/\rho$. With $P(s) = [P1(s); sP1(s); P3(s)]$ at low frequency, $|L(jw)| \approx |1 + j\gamma w||P1(jw)|/\sqrt{\rho}$.

Effects of the two design weights (matched to simulated Bode/step-response families in the slides):

- **Decreasing `rho`** (cheaper control) moves the Bode magnitude plot up -> higher loop gain -> faster response, less sensitivity to disturbances, but at the cost of more control effort (intuition: "input becomes cheaper").
- **Increasing `gamma`** (penalizing roll rate more) increases the phase margin -> less overshoot but slower response (intuition: increasing the cost of angular velocity discourages fast maneuvers).

2.9 Luenberger observer worked example (same roll-dynamics system)

MATLAB-style snippet shown:

```
R = 1;
G = [1 0 0; 0 .3 0]; Q = G'*G;
K = lqr(A,B,Q,R);             % K = [-1.0000, -0.4105, 0.1526]
damp(A-B*K)                   % closed-loop eigenvalues/damping/freq listed
pCL = eig(A-B*K);
L = place(A',C',pCL)';        % choose same poles for the observer (duality)
esR = estim(ss(A,B,C,0), L, 1, 1);   % build the observer/estimator object
```

Closed-loop LQR poles: a complex pair $-4.40 \pm 0.905i$ (damping ≈0.979, freq ≈4.49 rad/s) and a fast real pole `-49.6`. Observer gain obtained by placing the *same* pole locations via the transposed (dual) problem: $L = [7.6275, 29.0897, 37.9793]^T$.

A Simulink block-diagram slide contrasts **"LQR ideal feedback"** (full-state feedback using the true state `x`, i.e. `-Kx`) against **"LQR + observer feedback"** (feedback using the estimated state `xhat` from a noisy sensor, i.e. $-Kxhat$). Simulation plots (step response) show: (1) outputs — the observer-based response tracks the ideal-feedback response closely after a brief transient; (2) measurement — a noisy sensor signal converging to -1; (3) control input — a transient spike near t≈1 (when the step hits) then settling to a small noisy value around 0, demonstrating the observer-based controller performs comparably to ideal state feedback despite only having noisy output measurements.

3 Key equations reference

- Quadrotor translational/rotational dynamics: $m\ddot{p} = -Tr3 + mge3$, $\dot{R} = RS(\omega)$, $J\dot{\omega} = -S(\omega)J\omega + n$ — the plant being controlled throughout both chapters.
- Hierarchical control step 1: $u_T = -(T/m)r3d \Rightarrow T = m\|u_T\|$, $r3d = -u_T/\|u_T\|$ — decouples thrust magnitude/direction from the rotational subproblem.
- Position PD/PID law: $u_T = -kPe - kD\dot{e}[-kIIntegral(e)] - g + \ddot{p}$ — the position-loop control law; integral term needed for zero steady-state error under constant wind.
- Damping ratio from gains: $\xi = kD/(2\sqrt{kP})$ — lets you pick `(kP,kD)` for a target overshoot/settling-time spec.
- Phase-margin time-delay bound: $tau_c = PM/wc$ — max pure delay the loop tolerates before instability.
- Steady-state wind offset (PD only): $y(t) = r(t) + w0/kP$ — quantifies why integral action is needed.
- Attitude reference extraction: $[\cos(phi_d)\sin(theta_d), -\sin(phi_d), \cos(phi_d)\cos(theta_d)]$ $Rz(-\psi)r3d$ — converts a desired thrust direction into desired roll/pitch given desired yaw.
- Controllability matrix: $C = [B, AB, ..., A^{n-1}B]$, full rank `n` $\iff$ controllable — decides whether pole placement/LQR is achievable.
- Observability matrix: $O = [C; CA; ...; CA^{n-1}]$, full rank `n` $\iff$ observable — decides whether a convergent observer exists.
- Pole placement theorem: $\det(\lambda I - A + BK) = p(\lambda)$ solvable for any monic `p` iff `(A,B)` controllable.
- Luenberger observer: $x\dot{hat} = Axhat + Bu + L(y - Cxhat)$; error dynamics $xti\dot{l}de = (A - LC)xtilde$.
- Duality: eigenvalues of `A-LC` = eigenvalues of $A^T - C^T L^T \Rightarrow$ observer design = transposed controller design ($A \leftrightarrow A^T, B \leftrightarrow C^T, K \leftrightarrow L^T$).
- Separation principle: eigenvalues of the combined state-feedback+observer system = eig(A-BK) ∪ eig(A-LC) — justifies designing `K` and `L` independently (LTI only).
- Algebraic Riccati equation: $A^T P + PA + Q - PBR^-1B^T P = 0$, optimal gain $K = R^-1B^T P$, minimizing $J = Integral(x^T Qx + u^T Ru)dt$, min cost $= x^T(0)Px(0)$.
- Bryson's rule: $Q_i i = 1/(maxacceptablex_i^2)$, $R_i i = 1/(maxacceptableu_i^2)$ — practical starting point for LQR weight tuning.

- Kalman's equality: $|1 + L(jw)|^2 = 1 + \|GP(jw)\|^2/\rho$ — links the LQR weighting choice directly to open-loop Bode magnitude/phase margin.

4 Open questions / unclear points

- Ch3 leaves an explicit **unsolved exercise**: redo the wind-disturbance-rejection analysis with the disturbance entering as a velocity input rather than an acceleration input.
- The exact numeric values of `kw` (`k_omega`) and `kl` (`k_lambda`) used in the Ch3 slide 43-44 inner-loop pole-placement example are referenced (`roots([1,kw,kl])`) but not stated explicitly on the slides read — only the resulting 5-element gain vector is given.
- Ch4's $damp(A - BK)$ MATLAB output and the observer gain `L` are shown as pre-computed results (presumably from a live MATLAB/Simulink demo); the underlying `.m`/`.slx` files are not included in this PDF, only screenshots of the console/Simulink model.
- The `estim(ss(A,B,C,0), L, 1, 1)` call's last two arguments (sensor/noise indices) are shown without explanation in the slide — standard MATLAB Control System Toolbox usage, but not elaborated in the lecture.
- Ch4 defers the "optimal" version of the observer (Kalman filter) with "more on this later" — that material is presumably covered in Ch5/Ch6 (Sensors and State Estimation), not in this file.